

Future Stories (FSCE)-Style Sample Paper — Adventure & Beacon Maths

Time allowed: 45 minutes**Total marks:** 40**Questions:** 40**Calculator:** Not allowed

Name: _____

Date: _____

Instructions

Answer every question. Tick the box next to the answer you think is correct.

The number in square brackets [] shows the marks for each question.

Show your working in the boxes provided. You may not use a calculator.

Diagrams are not always drawn to scale.

- 1** What number does the Roman numeral XLVII represent? [1]

☐ A 47 ☐ B 57
☐ C 43 ☐ D 52

- 2** One January morning, four locations record these temperatures: Riverside -6°C , Hilltop -2°C , Valley 3°C , Lakeside -9°C . Which location recorded the coldest temperature? [1]

MORNING TEMPERATURES



☐ A Lakeside ☐ B Riverside
☐ C Hilltop ☐ D Valley

- 3** Round 68,472 to the nearest thousand. [1]

ROUNDING TO THE NEAREST 1,000



☐ A 68000 ☐ B 69000
☐ C 68500 ☐ D 68400

- 4** In the number 803,516, what is the value of the digit 5? [1]

PLACE VALUE OF 803,516

100,000s	10,000s	1,000s	100s	10s	1s
8	0	3	5	1	6

☐ A 500 ☐ B 5000
☐ C 50 ☐ D 5

- 5** At 6am the temperature was -4°C . By midday it had risen by 11°C . What was the temperature at midday? [1]

☐ A 7°C ☐ B 15°C
☐ C -7°C ☐ D 6°C

- 6** A school gate has the founding year carved into the stonework as MCMXLVIII. In which year was the school founded? [1]

☐ A 1948 ☐ B 1947
☐ C 1958 ☐ D 1848

Working

- 7** Work out 236×34 . [1]

☐ A 8024 ☐ B 7994
☐ C 8124 ☐ D 8004

- 8 A youth club has 187 pupils to transport in minibuses that each hold 12 pupils. What is the smallest number of minibuses needed so that every pupil has a seat? [1]

(A) 16 (B) 15
(C) 17 (D) 14

Working

- 9 Work out $8 + 6 \times (9 - 4)$. [1]

(A) 38 (B) 70
(C) 34 (D) 42

- 10 Three classes collected paper for recycling over a week, as shown in the table. The school's target was 400 kg. By how many kilograms did the three classes miss the target? [1]

PAPER COLLECTED THIS WEEK

Class	Paper collected
Class A	128 kg
Class B	96 kg
Class C	154 kg

(A) 22 kg (B) 18 kg
(C) 26 kg (D) 32 kg

Working

- 11 Which of these numbers is a factor of 84? [1]

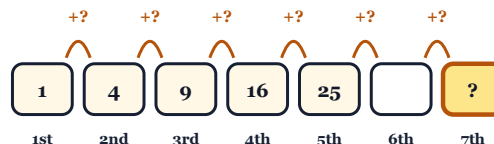
(A) 14 (B) 9
(C) 16 (D) 25

- 12 Which of these numbers is a prime number? [1]

(A) 61 (B) 51
(C) 57 (D) 63

- 13 The sequence of square numbers begins 1, 4, 9, 16, 25, ... What is the 7th square number? [1]

SQUARE NUMBERS



(A) 49 (B) 36
(C) 42 (D) 64

- 14 What is the value of 5^3 (5 cubed)? [1]

(A) 125 (B) 15
(C) 25 (D) 10

- 15 Which fraction is equivalent to $3/4$? [1]

THE FRACTION $3/4$



4 equal parts

(A) $9/12$ (B) $6/9$
(C) $8/10$ (D) $5/8$

- 16 A charity fun run is 15 km long. A runner has completed $2/5$ of the distance. How many kilometres has she run? [1]

THE FRACTION $2/5$



5 equal parts

(A) 6 km (B) 9 km
(C) 3 km (D) 10 km

- 17 Convert $7/20$ to a decimal. [1]

(A) 0.35 (B) 0.7
(C) 0.135 (D) 3.5

- 18** A stadium has 25% of its 3,200 seats reserved for members. How many seats are reserved for members? [1]

A 800 **B** 750
C 850 **D** 900

- 19** Which of these is the smallest: $\frac{3}{8}$, $\frac{2}{5}$, $\frac{1}{3}$, or 0.3? [1]

A 0.3 **B** $\frac{3}{8}$
C $\frac{2}{5}$ **D** $\frac{1}{3}$

Working

- 20** Work out $\frac{1}{4} + \frac{1}{6}$, giving your answer in its simplest form. [1]

A $\frac{5}{12}$ **B** $\frac{2}{10}$
C $\frac{1}{5}$ **D** $\frac{7}{12}$

- 21** A bike originally costs £120. In a sale it is reduced by 15%. What is the sale price? [1]

A £102 **B** £105
C £18 **D** £96

Working

- 22** Round 4.267 to 1 decimal place. [1]

A 4.3 **B** 4.2
C 4.27 **D** 4.1

- 23** Simplify the ratio 18 : 24 to its simplest form. [1]

A 3 : 4 **B** 6 : 8
C 9 : 12 **D** 2 : 3

- 24** Joe and Priya share £360 in the ratio 4 : 5. [1]
How much does Priya, who has the larger share, receive?

JOE : PRIYA

Joe



Priya



A £200 **B** £160
C £180 **D** £220

- 25** A map has a scale of 1 cm : 5 km. The distance between two villages on the map is 6.4 cm. What is the real distance between the villages? [1]

A 32 km **B** 30 km
C 11.4 km **D** 64 km

- 26** A pancake recipe for 4 people uses 200 g of flour. How much flour is needed to make pancakes for 10 people? [1]

A 500 g **B** 450 g
C 400 g **D** 550 g

- 27** In a school choir, the ratio of sopranos to altos is 3 : 2, and the ratio of altos to basses is 4 : 3. If there are 8 altos, how many basses are there? [1]

A 6 **B** 8
C 4 **D** 10

Working

- 28** Three angles lie on a straight line. Two of them measure 55° and 68° . What is the size of the third angle? [1]

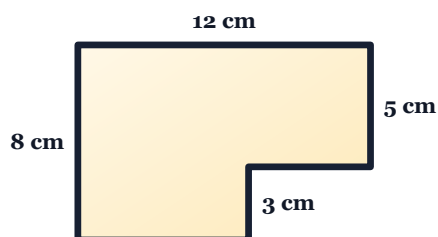
A 57° **B** 65°
C 62° **D** 50°

- 29** A triangle has one angle of 90° and another of 34° . What is the size of the third angle? [1]

A 56° **B** 46°
C 50° **D** 60°

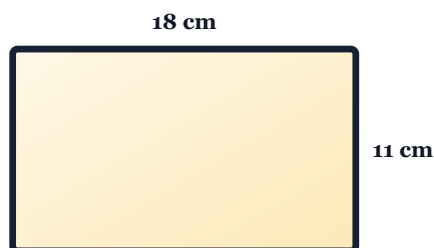
- 30 The diagram shows an L-shaped garden. The outer width is 12 m and outer height is 8 m, with a 5 m by 3 m rectangle cut from the bottom-right corner. What is the perimeter of the garden? [1]

L-SHAPED GARDEN



- 31 A rectangular sports hall floor measures 18 m by 11 m. What is its area? [1]

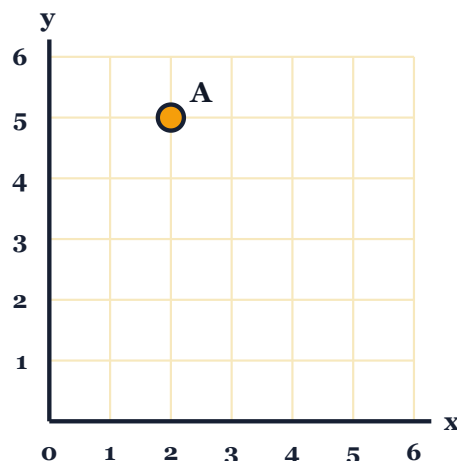
SPORTS HALL FLOOR



- 32 Which of these is NOT a property of a cube? [1]
- 33 How many lines of symmetry does a regular hexagon have? [1]

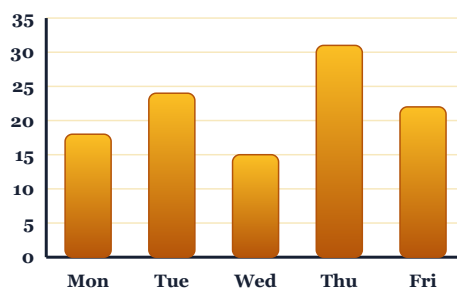
- 34 Point P is plotted on the grid at (2, 5). Point Q is 3 units to the right and 2 units down from P. What are the coordinates of Q? [1]

POINT P



- 35 The bar chart shows the number of books borrowed from the school library each day one week. How many more books were borrowed on Thursday than on Monday? [1]

BOOKS BORROWED THIS WEEK



- 34 [A] (5, 3) [B] (5, 7)
[C] (2, 3) [D] (5, 2)
- 35 [A] 13 [B] 12
[C] 9 [D] 16

- 36 The table shows how pupils in Year 5 travel to school. How many more pupils walk than travel by bus? [1]

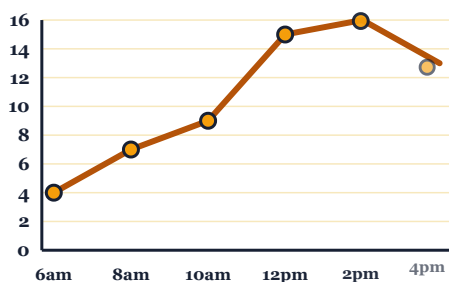
HOW YEAR 5 TRAVEL TO SCHOOL

Method	Number of pupils
Walk	34
Car	21
Bus	19
Bicycle	16

- 37 Five friends timed how long it took them to solve a puzzle: 8 minutes, 11 minutes, 6 minutes, 13 minutes and 12 minutes. What was their mean (average) time? [1]
- ☐ A 15 ☐ B 13
☐ C 18 ☐ D 21
- ☐ A 10 minutes ☐ B 11 minutes
☐ C 9 minutes ☐ D 12 minutes

- 38 The line graph shows the temperature recorded at a weather station every 2 hours during one day. Between which two times did the temperature rise the most? [1]

WEATHER STATION TEMPERATURE



- ☐ A 10am and 12pm ☐ B 6am and 8am
☐ C 12pm and 2pm ☐ D 2pm and 4pm

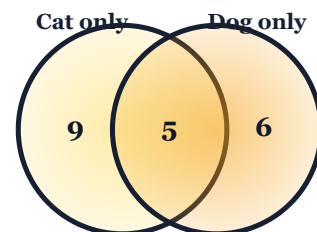
- 39 A school trip coach leaves at 09:45 and arrives 2 hours and 50 minutes later. At what time does it arrive? [1]

DEPARTURE TIME



- ☐ A 12:35 ☐ B 12:25
☐ C 11:35 ☐ D 12:45
- 40 In a class of 30 pupils, 9 have a pet cat only, 6 have a pet dog only, and 5 have both a cat and a dog. How many pupils have neither a cat nor a dog? [1]

PETS IN THE CLASS



- ☐ A 10 ☐ B 8
☐ C 11 ☐ D 14

Working